

**EFFECTS OF HORMONE ON GONADAL DEVELOPMENT OF *Sperata*
*aor***

A Thesis

By

Md. Afsar Ahmed Sumon

Examination Roll No. 1402030504, Reg. No. 0994

Session: 2009-10, Semester: July-December 2015

**MASTER OF SCIENCE
IN
FISH BIOLOGY AND GENETICS**



**DEPARTMENT OF FISH BIOLOGY AND GENETICS
FACULTY OF FISHERIES
SYLHET AGRICULTURAL UNIVERSITY, SYLHET-3100**

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Dedicated To

MY UNCLE, MD. ABUL KALAM AZAD

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ABBREVIATIONS AND ACCRONYMS

BW	Body weight
DoF	Department of Fisheries
EP	Early perinucleolar
EYG	Early yolk granule
FAO	Food and Agricultural Organization
GDP	Gross domestic production
GW	Gonad weight
H&E	Haematoxyline and Eosin
IUCN	International Union of Conservation of Nature
CF	Condition factor
LP	Late perinucleolar
LYG	Late yolk granule
M	Mature
N	Nucleus
OW	Ovary weight
PM	Premature
PG	Pituitary gland
SC	Spermatocytes
ST	Spermatid
SZ	Spermatozoa
TL	Total length
YV	Yolk vesicle

Effects of Hormone on Gonadal Development of *Sperata aor*

ABSTRACT

Two experiments were conducted to develop suitable artificial breeding tanks and breeding techniques of a threatened catfish, *Sperata aor* for a period of seven (7) months from May to November, 2015. The first experiment was conducted in cemented tanks with 3 different treatments viz. cemented floor only, cemented floor with 6-inch muddy bottom and cemented floor with 6-inch sandy bottom to determine their survivality in artificial condition. Each treatment was done in two tanks with same condition. The higher survival rate was observed 93.75% in cemented tank with sandy bottom followed by 25% in cement tank with muddy bottom and 6.25% in cemented tank only. Then cemented tank with sandy bottom condition was selected as breeding tank. In second experiment, two inducing agents viz. Flash hormone ((Synthetic gonadotropin releasing hormone analog (0.002%), Domoperidin (0.998%) and Prepelin glycol (99%)) and PG (Pituitary Gland) extracts were trialed for three different treatments with three replications for artificial breeding. To observe the effective dose for induced breeding, the females were first injected with Flash at the rate of 0.5 (T₁), 1.0 (T₂), and 1.5 (T₃) ml/kg body wt. while the second doses were performed with Flash at the rate of 1.0 (T₁), 1.5 (T₂), and 2.0 (T₃), ml/kg body wt. after 6 h from the first dose. On the other hand, the males were administered a single dose of 1 ml Flash /kg body wt. at the time of second dose of the female. Consequently, the females were first injected at the rate of 1.0 (T₁), 1.5 (T₂), and 2.0 (T₃) mg PG/kg body wt. while the second doses were performed with PG at the rate of 1.5 (T₁), 2.0 (T₂), and 2.5 (T₃), mg/kg body wt. after 6 hrs. from the first dose. Instead, the males were administered a single dose of 2 mg PG /kg body wt. at the time of second dose of the female. None of the trialed fishes released any eggs nor do any sperm by 72 hrs. after injection with different doses. Then the injected brood fishes were sacrificed to know the effectiveness of two different hormones. Histological study of gonad of *S. aor* indicated the presence of fully maturation stages of both male and female treated with Flash at T₂ treatment. It was full development stages in both ovaries and testis found in Flash hormone at T₂. These findings indicated that Flash hormone seems to be highly effective in gonadal maturation. Even though treated *S. aor* with Flash hormone did not response to release eggs and sperms but it is obvious that this hormone might be used in effectively to maximize the ovulation of *S. aor* in hatchery.

CHAPTER ONE

INTRODUCTION

Fish and fisheries have always played a dominant role in the economy, culture and tradition of the people of Bangladesh. In this country, fisheries sector has an important and potential contribution in the agro-based economic development, poverty elevation, employment generation, supplying of animal protein and earning of foreign currency. Bangladesh is a country enriched with water resources such as rivers, canals, *haors*, *baors*, *beels*, lake, etc. which are highly diversified.

Bangladesh is occupied in a riverine country in the South-east Asian region. Our country is vastly related in aquaculture development in terms of water resource management. Fisheries play a vital role in the development of our economy. Fish is also an important source of protein in Bangladesh and it alone supply 60% of total animal protein of the country. In 2013-14, fish production of Bangladesh was 3.548 million MT (DoF, 2014). Besides, aquaculture fish production of Bangladesh is the fourth highest in the world after China, India, Vietnam and capture fisheries production of Bangladesh is the sixteenth highest in the world (FAO, 2015). Moreover, in 2011-12 fisheries sector contributed 4.37% to national GDP, 23.37% to the agricultural GDP and 2.09% to foreign exchange earnings by exporting fish and fishery products (DoF, 2014).

The north-east part of Bangladesh i.e. the Sylhet basin is an important habitat for freshwater aquatic organisms. Sunamgonj, Sylhet, Moulvibazar, Kishorgonj, Netrokona and Mymensingh are the regions under Sylhet basin. Sylhet basin contains many kinds of freshwater habitats such as rivers, canals, *haors*, *baors*, *beels*, lake, etc.

Sperata aor has been considered as one of the most admired edible fishes among indigenous catfish species due to good taste and high market demand. The fish is carnivorous and subsists on various types of organisms such as fish, frogs, snakes, insects, earthworms, tadpoles, crustaceans and debris (Rahman, 2005).

According to IUCN (2003), 54 fish species of inland water of Bangladesh are now in endangered to critically endangered situation. *S. aor* is one of them which is now in critically endangered situation. *S. aor* is popularly known as ‘Ayre’ in Bangladesh and also in India. It is a tasty fish demanding comparatively high price in the market. Devi *et al.* (1991) mentioned

that it is an important commercial catfish of Aligarh fish market forming 8 to 12% of the total catch of catfish. Once the abundance of *S. aor* was very rich in Afghanistan, Pakistan, India, Nepal, Bangladesh and Myanmar. But its abundance has been reported to be declining day by day due to over exploitation and various ecological changes in our natural aquatic ecosystem. Our open water capture fisheries are under great stress and its sustainability is under danger because of changing aquatic ecosystem due to soil erosion. Dubey (1998) categorized *Sperata* sp. among the declining indigenous fauna of India.

Fries of *S. aor* were found during July-August in the pond with artificial nests, but not found in the pond with no artificial nests. The information reported in this study contributes to our knowledge on gonadal development of *S. aor* and enhances our understanding on its reproductive biology, which is useful for its aquaculture production and management. The fecundity was measured by gravimetric method which was found to vary from 59255.17 ± 184.03 to 70048.86 ± 131.99 . The regression equation for the relationships between log of fecundity and log of total length was $\text{Log}_{10} Y = 1.830 \text{Log}_{10} X + 1.717$, between log of fecundity and log of body weight was $\text{Log}_{10} Y = 0.894 \text{Log}_{10} X + 2.133$ and between fecundity and ovary weight was $\text{Log}_{10} Y = 0.394 \text{Log}_{10} X + 4.250$. The above equations showed linear relationship. A highly ($p < 0.01$) significant linear relationship was found to exist between fecundity and gonad weight. (Jabed, 2013).

The body surface of the male brood fish was inflamed, reddish and secreted a thick milky white fluid or scum from the body surface during breeding season (July-August). During breeding season (July-August) the abdomen of the mature female fishes were rounded and swollen, and the genital opening is reddish, rounded and protruded. Female obtained bigger size than the male partner in breeding season. (Afzal, 2014).

This fish has not been brought under aquaculture due to difficulties in spawning, breeding and rearing of brood stocks in captive conditions. But it is performed in group of asynchronous in spawning season. So, it is possible to produce in this species to long time in hatchery. To establish the induced breeding technique and culture in controlled environments of a riverine catfish, it is essential to know about its breeding biology and season. Unfortunately, no remarkable work has been done regarding its culture and breeding techniques (Jabed, 2013). But it needs artificial breeding to supply their fries in the culture system. But it needs some prerequisites for artificial breeding of ayre. By taking all above into consideration the present

study was set out to address some prerequisites for artificial breeding of *S. aor* with the following objectives:

- To develop suitable condition in the cistern for artificial breeding;
- To determine the effective dose of different hormones on gonadal development; and
- To investigate the gonadal maturity of fish injected with different inducing agents by histological observation.

CHAPTER TWO

REVIEW OF LITERATURE

Although culture of catfishes has been well established in the Indian sub-continent but *S. aor* has received very little attention for this purposes. There is a dearth of information on artificially survivality, artificial breeding and effective doses of hormones of *S. aor*. However, related literatures of the proposed work are reviewed below:

Afzal (2014) indicated that ovaries of *S. aor* were paired, more or less elongated. The color of ovary varied from reddish-brown to light yellowish during breeding season of *S. aor*. The GSI values were gradually rising in the month of July to August and the oocyte development of *S. aor* is an asynchronous pattern, so fish breed in throughout the year.

Jabed (2013) observed that the spawning season of *s. aor* extended from May-August and *S. aor* has group asynchronous mode of ovarian development and is capable of spawning several times in a year under favourable conditions and found ovary of *S. aor* indicated the presence of four developmental stages: a) Early perinucleolar stage oocytes b) Late perinucleolar stage oocytes c) Yolk vesicle stage oocytes d) Yolk granular stage oocytes. These findings indicated that *S. aor* has group asynchronous mode of ovarian development and is capable of spawning several times in a year under favourable conditions. Again of histological study of testis indicated the presence of spermatocyte (SC), spermatid (ST) and spermatozoa (SZ) stages.

Ulka (2011) studied the peak breeding season and gonadal maturity stages of native, Thai, and hybrid koi (*Anabas testudineus*) during the breeding season (April- June) and found the highest value of GSI for native male was 1.43 ± 0.68 in June and the lowest was 0.97 ± 0.16 in May and for female it was highest in May 9.16 ± 12.23 and lowest in June 3.01 ± 0.3 .

Akhter (2011) stated that *S. seenghala* is one of the important bagrid catfishes and was once abundantly available in Bangladesh and found early and developing germ cells were mostly observed during April- May; gradually maturing in June; and mature germ cells during July-August sections of testes.

Maya (2011) identified the processes of gametogenesis in a threatened fish species *Mystus cavasius* from the Mymensingh region and mature stages of oocytes (PM and M) were observed from early July to August samples and developing oocytes (O, PVO and PNO) were mostly observed in the September samples.

Gaikwad et al. (2009) studied on maturity of oocyte (egg) in *Channa gachua*. Development of oocytes was measured with the help of oculometer and maturity stage of gonad found varies according to different breeding phases. More number (30.7%) of oocytes (eggs), observed in pre-spawning and spawning period, whereas less number (0.99%) oocytes (eggs), observed post-spawning period.

Nasrin (2007) studied the reproductive biology of *Rita rita* and found that the highest GSI of 0.41 ± 0.10 in June with the highest fecundity ($58,760 \pm 2,809$) and presumed that the fish may breed between July and August.

Ziki (2006) studied the reproductive biology of *Mystus vittatus* of a perennial water body in Bangladesh. He suggested that the spawning season of *M. vittatus* were from February to July. Nine stages of oocyte development were observed and indicated that *M. vittatus* was single spawner, and its spawning frequency or strategy was synchronous type.

Moniruzzaman (2006) studied the reproductive biology of Indian river shad *Gudusia chapra* from a perennial water body in Bangladesh and the author suggested that the spawning seasons of *G. chapra* were from March to May and July to September.

Kunda et al. (2004) studied the length-weight relationship, condition factor and relative condition factor of *Barbodes gonionotus*, *amblypharyngodon mola* and *Cyprinus carpio*. Length-weight relationship of *B. gonionotus*, *A. mola*, *C. carpio* were established as $\text{LogW} = -1.2286 + 2.5007 \log L$; $\text{LogW} = -1.6521 + 2.5331 \log L$ and $\text{LogW} = -1.8940 + 3.0829 \log L$, respectively. The coefficient of correlation (R) was found to be 0.996, 0.997 and 0.998 for *B. gonionotus*, *A. mola*, *C. carpio* respectively.

Sarker et al. (2002) investigated the fecundity and gonadosomatic index of *Mystus gulio* from March to December and concluded that there was a wide spawning season of the species the peak being in July. Fecundity of the fish increased with the increase in size, weight gonad weight as well.

Rheman et al. (2002) observed the fecundity and gonadosomatic index (GSI) of *Liza parsia* (Ham.) and the fish was found to spawn for several months with two spawning peaks. One was in December and the other was in February as indicated by two peaks of the gonadosomatic index. The fecundity of the fish ranged from 193443 to 301700.

Afroz (2000) studied breeding biology of *Gudusia chapra* and reported that the fish has two breeding seasons in a year from March to April and from August to September. The minimum size at maturity was 111 mm for the male and 136 mm for female.

Afroz et al. (1999) estimated fecundity of *Gudusia chapra*, which varied from 1105 to 26532. The relationship between fecundity, total length, total weight and gonadal weight were calculated and a positive relationship was found to exist between the fecundity and gonadal weight.

Narejo et al. (1998) reported reproductive biology of *Tenualosa ilisha* and found that progressive increase in ova diameter up to 1.24 mm in July-August as a spawning season of the fish. The number of ova per gram of ovary weight was 6,363. The maximum fecundity was calculated as 11,56,383 from a fish measuring 480 mm (TL) and the minimum being 64,680 from a fish measuring 201 mm (TL). The fecundity data was plotted against total length, body weight and growth of the fish. The fecundity-growth relationship gives better information as compared to other variables.

Faruq et al. (1996) calculated fecundity and the relationship of fecundity with length and weight of *Clarius batrachus*. He reported 3,028 eggs from a fish of total length 18.1 cm and 9,064 eggs from a fish of total length 26.2 cm. the average number of ova per gram of ovary weight was 362. The ova diameter was found to be 810 μm .

Palmer et al. (1995) studied the ovarian development in freshwater drum, *Aplodinotus grunnius*. They pointed out that the ovaries contained a continuum of oocyte sizes and the largest oocytes were only ovulated at spawning. They also reported that a large percentage of the oocytes became atretic mid-way through the spawning season, while GSI were still high and that might have reduced the spawning potential of Red Lake drum fish.

Shaima et al. (1992) reported the size frequency distribution in a year cycle and fecundity of *Mystus tengara* during its peak breeding season. The fecundity varied from minimum 720 to maximum 5233 eggs.

Khan et al. (1990) studied the reproductive biology of a tropical catfish, *Mystus nemurus* (Cuvier and Valenciennes) and reported that in female, GSI from September to April were not significantly different among months. Mean GSI from February and November were insignificant ($P>0.05$) than in May, August, October and November. Fecundity increased

linearly with length and weight and ranged from 6,900 in a fish of body length 34.8 cm to 93,510 in a fish of 45 cm.

Barua *et al.* (1987) studied on the spawning behaviour of *Mystus tengara* (Hamilton). They reported that *Mystus tengara* could be breed by frog pituitary injection. Spawning activity started 4 hours' post-injection of two pituitary glands and ended 6-7-hour post-injection resulting in the spawning of fish.

Mollah (1986) studied the cyclic changes in the ovary of freshwater catfish, *Clarias macrocephalus* (Gunther) and observed seven stages: oogonial stage, chromatin nucleolar stage, early perinucleolar stage, late perinucleolar stage, yolk-vesicle stage, early yolk-granule stage and late yolk-granule stage. Oogonial multiplication, formation of new oocytes and transformation into primary growth- phase oocytes was observed in June and July and vitellogenesis from August to September.

CHAPTER THREE

MATERIALS AND METHODS

Methodology is an indispensable and important part of scientific research. Appropriate methodology is a prime necessity that enable the researchers to collect valid and reliable information and to analysis the information properly in order to achieve a good conclusion.

3.1 Study area

The study was conducted in the earthen pond, histology lab of Fish Biology and Genetics Department and mini fish hatchery complex of the Sylhet Agricultural University, Sylhet.

3.2 Experimental design

3.2.1 Selection of suitable artificial brood tank

The experiment one was carried out with three treatments viz. T1 (Cemented tank), T2 (Cemented tank with 6-inch muddy bottom) and T3 (Cemented bottom with 6-inch sandy bottom). Each treatment was carried out with two replications (Fig. 3.1). The male and female brood of *S. aor* were stocked at 1:1 ratio. A market available pelleted feed named “mega fish feed” (grower) having Moisture 25%, Protein 26%, Lipid 5%, Carbohydrate 8%, Ash 18%, Calcium 1.8%, Phosphorus 0.6% was given at 5% body weight every day. Besides, some small fishes like male *tilapia*, *punti fries*, *mola etc.* were given in each tanks time to time as their live feed. All the tanks were with water flow through system with aeration. All the important water quality parameters were checked by using HACH kit. Fish were carefully monitored regularly whether there any lesion appear or not. A post mortem was done for every fish carefully.

3.2.2 Induced breeding

Cemented tanks with sandy bottom were selected for this experiment. Two different hormone trials viz. Flash hormone and PG (Pituitary Gland) hormone were used as inducing agents. Each trials having three treatments and every treatment was carried out with three replications (Fig 3.1). After hormone trial, injected fishes both males and females were kept in same tank at 1:1 ratio for 72 hrs. Then injected brood fishes were sacrificed to know the effectiveness of applied hormones and observed their histological changes of gonads.

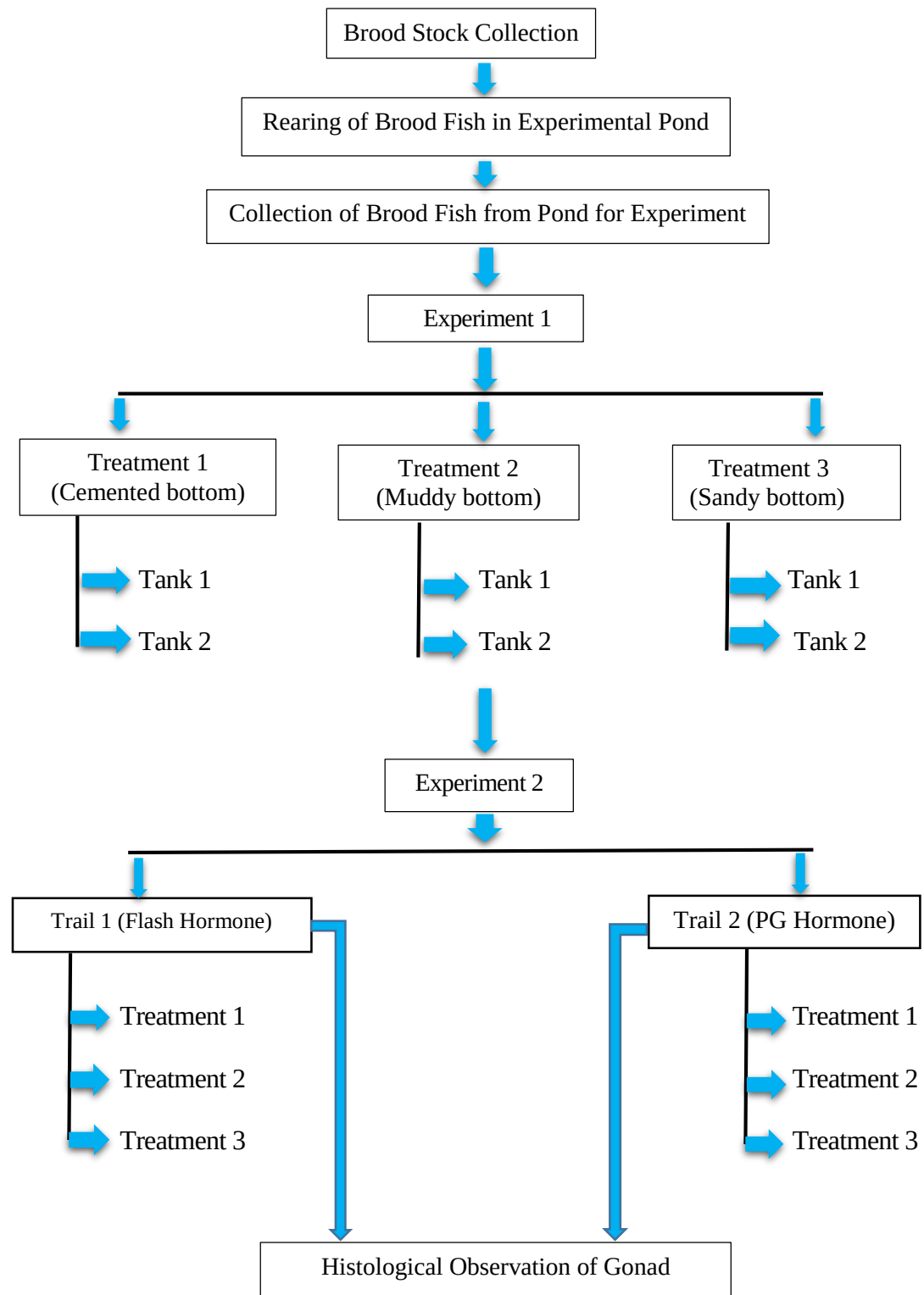


Fig. 3.1 Flow diagram of the research work

3.3 Brood stock Collection

Brood fish of *S. aor* were collected from American Fish Farm of Zakiganj Upazila under Sylhet district (Fig. 3.2) and live transported in plastic open mouth drum with orsaline. They were transported to the experimental pond complex at Sylhet Agricultural University and released into the experimental ponds carefully after acclimatization.

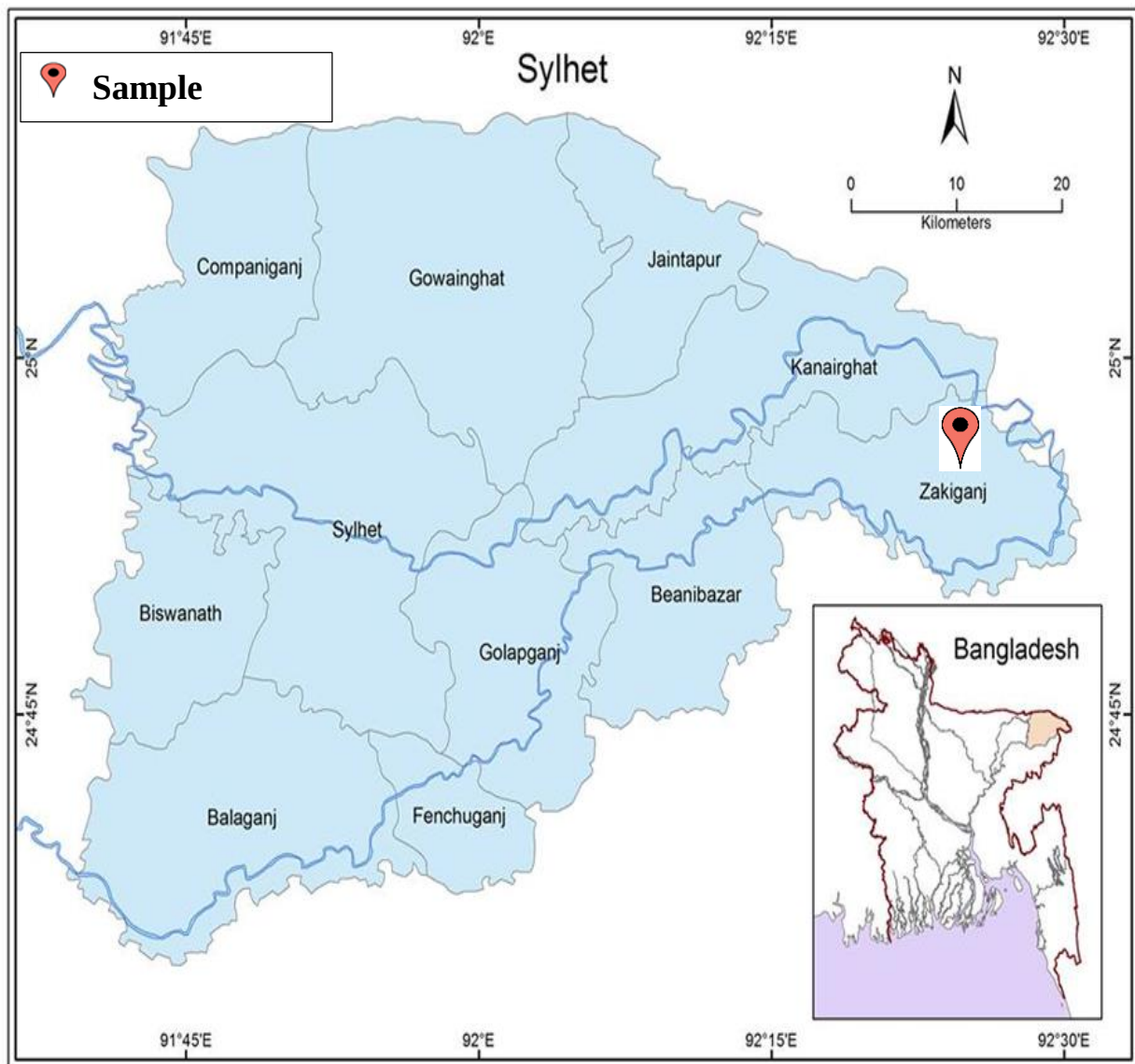


Fig. 3.2 Location of the brood stock collection and study area on the map

3.4 Rearing of brood fish

Collected *S. aor* broods were released in total 9 ponds at 25 no's/dm stocking density. A market available pelleted feed named "Mega fish feed" (grower) having Moisture 25%, Protein 26%, Lipid 5%, Carbohydrate 8%, Ash 18%, Calcium 1.8%, Phosphorus 0.6% were applied twice a day at the rate of 3-5% body weight. Besides, some small live fishes (*punti*, *tilapia fries*, *mola* etc.) were released in the same ponds as they fed them due to their carnivorous nature.



Fig. 3.3 Brood fish rearing site (mini hatchery pond complex of the Faculty of Fisheries, Sylhet Agricultural University, Sylhet)

3.5 Preparation of rearing tanks

Each tank of 4 m × 1.5 m × 1 m (length × width × height), was established within a flow-through system with aeration, with water being supplied from an overhead tank containing de-chlorinated water. Three sets of tanks were prepared with cemented floor with 6inch muddy bottom, cemented floor with 6inch sandy bottom and controlled without mud and sand.



Cemented bottom

Muddy bottom

Sandy bottom

Fig. 3.4 Preparation of tanks bottom by adding different materials

3.5.1 Collection and selection of brood fishes

The brood fish were collected from the brood tanks whenever necessary by reducing the water of the ponds. Sometimes cast net was used to collect brood fish. Good and healthy broods were selected for breeding. Male and female broods were selected on the basis of some external features (Jabed, 2013). Briefly, male broods were selected on the inflamed body surface and secreted a thick milky white fluid or scum from the body surface. On the other hand, mature female fishes were selected by observation rounded, swollen belly, and reddish protruded genital opening. Female obtained bigger size than the male partner in breeding season. (Afzal, 2014). Selected brood fishes were released into prepared experiment tanks.

3.5.2 Water quality parameters

The water quality parameters such as temperature, pH, dissolved oxygen (DO), alkalinity, ammonia, hardness and nitrite were monitored weekly throughout the experimental period. Water temperature, pH, dissolved oxygen (DO), alkalinity, ammonia, hardness and nitrite of the ponds was measured with the help of HACH kit. All analyses were done in the Laboratory of Fish Biology and Genetics of Sylhet Agricultural University, Sylhet.

3.6 Induced breeding trials

3.6.1. Breeding trial with Flash hormone

In this experiment, three doses of Flash hormone 0.5, 1.0, and 1.5 ml/kg body wt. considered as treatments T1, T2, and T3 respectively were used with three replications of each. For each treatment 6 males and 6 females were used. So 36 brood fishes were injected for induced breeding that composed of 18 females and 18 males. For dose optimization, maturation rate, histological examination and survival rate was determined.

3.6.1.1 Treatment 1

Three pairs of *S. aor* fish were treated with Flash hormone with three replications. First dose 0.5 ml/kg body wt. of Flash hormone was administered to the female at 06.00 h, at six hours' interval; second dose of 1.0 ml/kg body wt. of Flash hormone was given to female at 12.00 hours and first dose 1.0 ml/kg body wt. to male. Then treated fish were kept in the different plastic container which were subjected under water showering to induce the breeding.

3.6.1.2 Treatment 2

First dose 1.0 ml/kg body wt. of Flash hormone was administered to the female at 06.00 h, at six hours' interval; second dose of 1.5 ml/kg body wt. of Flash hormone was given to female at 12.00 h and first dose 1.0 ml/kg body wt. to male. Then treated fish were kept in the different plastic container which were subjected under water showering to induce the breeding.



Fig.3.5 Flash hormone.

3.6.1.3 Treatment 3

First dose 1.5 ml/kg body wt. of Flash hormone was administered to the female at 06.00 h, at six hours' interval; second dose of 2.0 ml/kg body wt. of Flash hormone was given to female at 12.00 h and first dose 1.0 ml/kg body wt. to male. Then treated fish were kept in the different plastic container which were subjected under water showering to induce the breeding.



Fig. 3.6 Injecting the fish with flash hormone and kept in plastic drum with continuous water circulation.

3.6.1.4. Hormone administration

After preparation of Flash solution, it was injected in brood fish. Fish was caught carefully by net and they were weighted by a balance separately. They were gripped softly by a cleaned towel. Flash was then injected near the pectoral fin base (Fig. 3.6). The amount of Flash solution for each fish was determined before according to the body weight of the repeated broods. The doses of injections are given in Table 1.

Table 1. Doses of Flash hormone for male and female broods of *S. aor*

Treatments	First dose for female (ml Flash/Kg body wt.)	Second dose for female (ml Flash/Kg body wt.)	Time interval (hours)	Single dose for male (ml Flash/Kg body wt.)
T ₁	0.5	1.0	6	1.0
T ₂	1.0	1.5	6	1.0
T ₃	1.5	2.0	6	1.0

3.6.2 Breeding trial with pituitary gland (PG)

In this experiment, three doses of PG hormone 1.0, 1.5, and 2.0 mg/kg body wt. considered as treatments T₁, T₂, and T₃, respectively were used with three replications of each. After selecting mature male and female PG extracts are used to induce them to breed.

3.5.2.1 Treatment 1

Three pairs of *S. aor* fish were treated with PG with three replications. First dose 1.0 mg/kg body wt. of PG hormone was administered to the female at 06.00 h, at six hours' interval; second dose of 1.5 mg/kg body wt. of PG hormone was given to female at 12.00 h and first dose 2.0 mg/kg body wt. to male. Then treated fish were kept in the different plastic container which were subjected under water showering to induce the breeding.

3.6.2.2 Treatment 2

First dose 1.5 mg/kg body wt. of PG hormone was administered to the female at 06.00 h, at six hours' interval; second dose of 2.0 mg/kg body wt. of PG hormone was given to female at 12.00 h and first dose 2.0 mg/kg body wt. to male. Then treated fish were kept in the different plastic container which were subjected under water showering to induce the breeding.

3.6.2.3 Treatment 3

First dose 2.0 mg/kg body wt. of PG hormone was administered to the female at 06.00 h, at six hours' interval; second dose of 2.5 mg/kg body wt. of PG hormone was given to female at 12.00 h and first dose 2.0 mg/kg body wt. to male. Then treated fish were kept in the different plastic container which were subjected under water showering to induce the breeding.

3.6.2.4 Hormone Administration

After preparation of PG solution, it was injected to brood fish. To inject PG, the brood fish were taken out from the cistern by scoop net and kept on foam. Then the head region of the brood fish was wrapped by a wet, soft cloth. Based on the body weight of the gravid male and female the required volume of extract was taken in a graduated 1ml hypodermic syringe. The extract was injected intramuscularly to the fish on the pectoral side above the lateral line (Figure 3.7). During injection needle was inserted at about 45° angle. The doses of injections are given in Table 2.



Fig. 3.7 Injecting the fish with PG hormone and kept in plastic drum with continuous water circulation.

Table 2. Doses of PG for male and female broods of *S. aor*

Treatments	First dose for female (mg PG/Kg body wt.)	Second dose for female (mg PG/Kg body wt.)	Time interval (hours)	Single dose for male (mg PG/Kg body wt.)
T ₁	1.0	1.5	6	2.0
T ₂	1.5	2.0	6	2.0
T ₃	2.0	2.5	6	2.0

3.7 Histology of gonad

After 72 hours post injection, the one pair of treated fish from each treatment were sacrificed to see the gonadal maturity by histological observation.

3.7.1 Ovary

The ovary of each fish was taken out very carefully and preserved in Carnoy's fluid with labeled vials for further study. The weight of the ovary was measured very carefully with the help of a sensitive portable electronic balance. The preserved gonads were taken out in a perforated plastic holder, which was covered by perforated steel plates. Cleaning, infiltration and dehydration process were done in an automatic tissue processor using a series of alcohol of increasing concentrations, three changes of xylene and finally through molten wax (three series) as mentioned in Table 1. Gonads of female fish collected from April to December 2015 were examined histologically to study the different maturation stages of ovary. Gonadal sections were examined and monthly developmental variations were observed. Paraffin embedded blocks were cut by microtome knife at 4-5 μ m size and left the sections into a water bath at a temperature of 40°C for cleaning. The sections were placed on a glass slide and kept overnight on a slide drier hot plate at a temperature of 78°C. Then the sections were stained routinely with haematoxyline and eosin (Ankley *et al.* 2001).

3.7.2 Testis

Small pieces of testicular tissue, measuring about 4-5 mm in length were removed from the mid- region of the testes and fixed in Carnoy's solution for 48h and subsequently processed for histology. Each piece of tissue, embedded in paraffin wax was sectioned at 5 μ m and stained routinely with haematoxyline and eosin.

3.7.3 Microscopic observation

The microscopic observation of gonad was studied by using of histological tools followed by standard protocol (Ankley *et al.* 2001). These steps are mentioned as below:

3.7.4 Fixation of gonad

The samples of gonad were cut to 0.5-1 cm thickness and preserved in glass jar with Carnoy's fixative for 24 hours at room temperature.

Table 3. Chemical composition of Carnoy's fixative

Serial No.	Chemical name	Amount
1.	Propanol	60 ml
2.	Chloroform	30 ml
3.	Glacial acetic acid	10 ml

3.7.5 Washing

Fixed tissues were washed in running tap water for 3 hours. Washing with running tap water helps to remove excess fixative from the tissue and reduce the color of fixative allowing better assessment of tissue content.

3.7.6 Dehydration

The samples that kept in 10% formalin fixative were started dehydration from 70% ethyl alcohol (step 1) and proceed on up to 100% ethyl alcohol (step 5). The dehydration schedule is given below:

Table 4. Protocol of Dehydration process on gonadal histology

Serial No.	Solution	Time
1	50% ethanol	3 hours
2	70% ethanol	3 hours
3	80% ethanol	Overnight
4	90% ethanol	3 hours
5	95% ethanol	3 hours
6	98% ethanol	3 hours
7	100% ethanol	3 hours
8	100% ethanol	Overnight

3.7.7 Clearing

Dehydration gonads were cleared by using Xylene twice successfully for 40 minutes to remove traces of alcohol in order to have consistent paraffin blocks. The process is as followed:

Table 5. Protocol of clearing process on gonadal histology

Serial No.	Solution	Time
1	Xylene	40 minutes
2	Xylene	40 minutes

3.7.8 Infiltration

In this step the gonadal samples were kept with liquid paraffin in glass jar inside Hot Air Oven at 70°C temperature which performs the following steps:

Table 6. Protocol of Infiltration process on gonadal histology

Serial No.	Solution	Time
1	Liquid paraffin	1-2 hours
2	Liquid paraffin	1-2 hours
3	Liquid paraffin	Overnight

3.7.9 Embedding

After infiltration, the tagged gonadal samples were kept in plastic small size cup separately and filled with melted paraffin. Then the embedded samples block was allowed to cool in room temperature for 1-2 days.



Fig. 3.8 Water Bath, oven and embedded samples block

3.7.10 Trimming

Trimming is a process in which the undesirable wax layers of the embedded blocks are trimmed by knife to obtain suitable blocks. Trimming was done by using old microtome blades. Trimming allowed easy sectioning. In this step both side trimming and surface trimming were conducted.

3.7.11 Sectioning

Paraffin embedded blocks was cut by microtome blade in microtome machine at 5µm size. The sections were placed on lower part of a glass slide previously tagged and filled with water drops. The sections were kept overnight at room temperature for proper drying. Then the sections were stained routinely with haematoxyline and eosin dye as per the schedule given below:

3.7.12 Staining

Staining is a process by which samples are stained with various dyes and staining materials so that their components are visible under microscope. The sections were stained routinely with haematoxyline and eosin as per the schedule given below:

Table 7. Standard staining procedure of Haematoxyline and Eosin

SL No.	Solution	Time	Process
1	Xylene	10-15 minutes	Clearing (1-2)
2	Xylene	10-15 minutes	
3	100% alcohol	2-5 minutes	Dehydration (3-8)
4	100% alcohol	2-5 minutes	
5	95% alcohol	2-5 minutes	
6	90% alcohol	2-5 minutes	
7	80% alcohol	2-5 minutes	
8	70% alcohol	2-5 minutes	
9	Normal water	2-5 minutes	Staining (9-14)
10	Distilled water	2-5 minutes	
11	Haematoxyline	5-10 minutes	
12	Wash in tap or running water	30 minutes	
13	Distilled water	2-3 minutes	
14	Eosin yellow	2-5 minutes	
15	70% alcohol	15 seconds	Rehydration (15-19)
16	80% alcohol	15 seconds	
17	90% alcohol	15 seconds	
18	95% alcohol	15 seconds	
19	100% alcohol	15 seconds	
20	100% alcohol	15 seconds	
21	Xylene	15 seconds	Clearing (21-22)
22	Xylene	15 seconds	

3.7.13 Mounting

Canada balsam was used for mounting as a mounting agent. A jot of Canada balsam is put on each slide followed by attachment of cover slip (22mm×22mm). After mounting the slides were put for several hours at room temperature.



Fig. 3.9 Sectioning by Microtome Machine (Slice cut4062), sections were stained routinely with hematoxyline and eosin, and mountain

3.7.14 Microscopic observation of gonads

After mounting the histological slides were observed under compound light microscope and photographed using an attached digital camera (AmScope, M158C-E Compound Monocular Microscope+ Digital camera USB imager, America). Total 10 histological slides were taken for this study. In the present study, the oocyte germ cells including the yolk vesicle (cortical alveoli), vitellogenic and maturation stages were identified according to the method described by Gentek et al. (2009).

3.7.15 Data analysis and Photography

The slides were observed with a light microscope and images were computer digitized using an image capture analysis system (AmScope, M158C-E Compound Monocular Microscope+ Digital camera USB imager, America)), to determine the histological stage development of ovarian oocyte and testes. All the morphometric data were transferred to log values. Photographs were taken by using a digital camera (Model no. SONY DSC-W220) and microscope with photograph facilities. All statistical analyses were carried out using IBM SPSS statistics software, version 20. results are presented as mean±standard deviation



**Fig. 3.10 An AmScope M158C-E Compound Monocular Microscope+ Digital camera
USB imager, America)**

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 Experiment one

4.1.1 Improvement of survival rate of *S. aor* in artificial condition

The catfish *S. aor* is a threatened species in Bangladesh (IUCN 2003). In rivers and estuaries, it prefers sandy and clear water. It is carnivorous and fed on insects, mollusks, shrimp and fishes. Their body is extremely slimy when captured. In natural condition, it builds nest among pebbles or in the mud pits on river bed. The breeding pair used to prepare a simple type of nest by excavating pits in the clayey and sandy exposed mud flats of the river behind a projecting slab or shelf of rock to protect the young from the current.

Until now fishery of *S. aor* mainly depends on capture from nature. This fish has not been brought under aquaculture due to difficulties in rearing of brood stocks in captive conditions as well as response to hypophysation. We normally use cemented tanks in hatchery, which seems not to prefer by *S. aor*. Unfortunately, no remarkable work has been done regarding its congenial conditions in hatchery. Therefore, the present study was undertaken to determine a suitable survival parameter to obtain maximum survival rate of *S. aor* in cemented tank in hatchery.

4.1.2 Treatment 1: Brood fish rearing in cemented tanks

In current study, two cemented tanks were prepared and brood fishes of *S. aor*, weighing 330-450 g total body weight (TBW) and 96-113 cm standard length (SL) were obtained from the pond complex and transported live to the fish hatchery tanks. Eight (8) fishes were distributed in each tank. The male and female brood of *S. aor* were stocked at 1:1 ratio. Fishes were carefully monitored regularly whether any changes of fishes appeared or not. Over one week later most of fishes found with broken and infected barbells, reddish and infected skin. Out of 8 fishes, 7 and 8 fishes were died in Tank 1 and Tank 2, respectively by 3 weeks (Table 9). It might be due to physical injury which occurred from frequent contact with hard bottom surface of the tanks. Because all the dead fishes seriously injured on abdominal sides. Besides, the skin and barbells of fishes were observed broken followed by gradual infections and lesions and thus higher mortality occurred. Only one infected fish was thriving after 3 weeks. The average percentage of survival rate was 6.25 in cemented tanks (Table 8).

Table 8. Measurement of survival rate in cemented tanks.

Tank No.	No. of Fish Release	No. of dead fish/Week			No. of Live Fish	Survival Rate (%) / Tank	Average Survival rate (%)
		1st	2nd	3rd			
1	8	0	2	5	1	12.5	6.25
2	8	1	3	3	0	0	



Fig.4.1 Infection caused by hard surface of cemented tanks

4.1.3 Treatment 2: Brood fish rearing in cemented tanks with muddy bottom

After completion of treatment 1, tanks were re-arranged having sides pasted with cement and bottom filled with mud. Live brood fishes of *S. aor*, weighing 350-470 g total body weight (TBW) and 97-117 cm standard Length (SL) were obtained from the pond complex and live transported to the prepared tanks of fish hatchery. They were disinfected with salt bath and acclimatized for 24 hours. The male and female brood of *S. aor* were stocked at 1:1 ratio. A total of 16 fish were released in two tanks each tank had 8 fishes. Fishes were routinely monitored to observe the condition of brood fishes. After four weeks of stocking, it was observed only two brood fishes survived in each tank and other brood fishes found deceased. The frequent contact with hard surface of cemented tank resulted lesions in fish body and turbidity caused oxygen deficiency might be accountable for the mortality in this treatment.

The average percentage of survival rate was found only 25 in cemented tanks with muddy bottom layer (Table 9). Catfishes are bottom dwelling fishes with large flattened heads, barbels, an adipose fin, mildly poisonous dorsal and pectoral spines but no scales. However, in treatment 2 brood fishes survived more days than treatment 1.

Table 9. Measurement of survival rate in cemented tanks with muddy bottom layer.

Tank No.	No. of Fish Release	No. of dead fish/Week				No. of Live Fish	Survival Rate (%) / Tank	Average Survival rate (%)
		1st	2nd	3rd	4th			
1	8	0	2	3	1	2	25	25
2	8	0	3	2	1	2	25	

4.1.4 Treatment 3: Brood fish rearing in cemented tanks with sandy bottom

In this treatment, tanks condition was again modified. Here bottom of the tanks was filled with sands and then tanks cleaned well by using disinfectant. Live brood fishes of *S. aor*, weighing 340-475 g total body weight (TBW) and 94-115 cm standard length (SL) were obtained from the pond complex and transported in the prepared tanks of fish hatchery. Brood fishes were stocked in tanks at 1:1 male and female ratio with 8 fishes per tank. Five weeks later, all brood fishes of tank 1 found alive with healthy condition, meanwhile, only one brood fish of tank 2 was found dead. Survival rate of brood fish were observed 100% and 87.5% in tank 1 and tank 2, respectively. While, cumulative survival rate of treatment 3 was estimated 93.75% (Table 10).

Table 10. Measurement of survival rate in cemented tanks with sandy bottom layer.

Tank No.	No. of Fish Release	No. of dead fish/Week					No. of Live Fish	Survival Rate (%) / Tank	Average Survival rate (%)
		1st	2nd	3rd	4th	5th			
1	8	0	0	0	0	0	8	100	93.75
2	8	0	1	0	0	0	7	87.5	

Sandy bottom of the experimental tanks might be the reason of low mortalities in the current treatment. Sandy bottom may allow *S. aor* not to abrasion on their pelvic sides. Hence, physical injuries were minimal here. These findings are in agreement with the findings of Ranganathan and Radha (1966) and Saigal (1982). They reported *S. aor* preferred sandy beds and clear water for living.

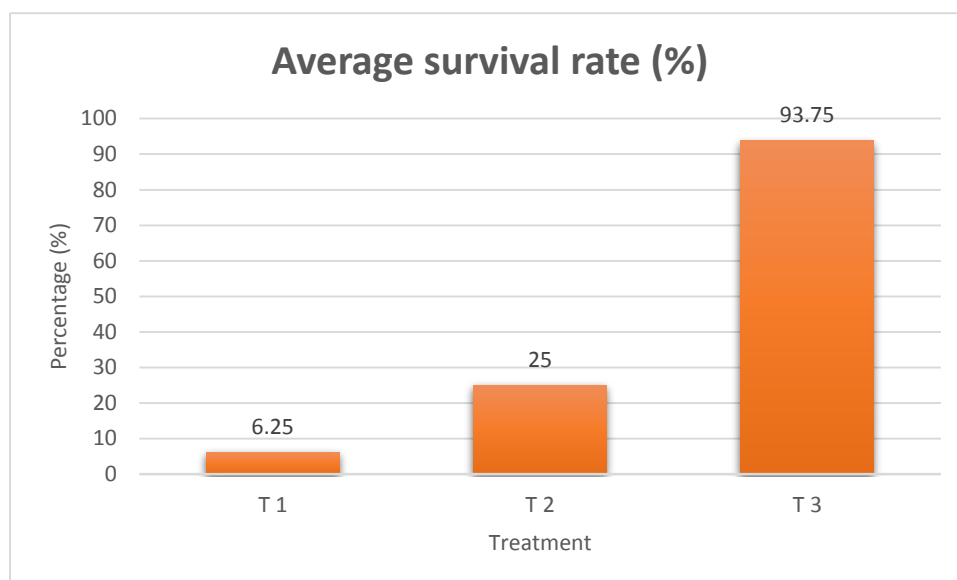


Fig.4.2. Average survival rate among three treatments

4.1.5 Water quality parameters

The water quality parameters such as temperature, pH, dissolved oxygen (DO), alkalinity, ammonia, hardness and nitrite were monitored weekly throughout the experimental period..Surface temperature was fluctuated from 28 °C to 30 °C, Dissolved oxygen was 7 to 8 mg/l, pH was 6.8 to 7.6, Turbidity was oscillated from 9 cm to 28 cm, Nitrite differed from 0 mg/l to 0.05 mg/l, Ammonia was observed from 0.011mg/l to 0.18 mg/l, Alkalinity was fluctuated from 68 mg/l to 80 mg/l, the value of hardness ranged from 60 mg/l to 89 mg/l.

Table 11. Mean±SD water quality parameters of different treatments of the study

Water Quality Parameters	T ₁	T ₂	T ₃
DO (mg/l)	7.5±0.71	7.5±0.71	7.5±0.71
pH	6.9±0.14	7.2±0.0	7.5±0.14
Temperature (° C)	27.5±0.71	29±1.41	27.5±0.71
Turbidity (cm)	20±2.83	27±1.41	9.5±0.71
Hardness	83.5±7.78	71.5±3.54	69±1.41
Nitrite (mg/l)	0.045±0.01	0.035±0.01	0.005±0.01
Ammonia (mg/l)	0.16±0.03	0.15±0.03	0.015±0.03
Alkalinity (mg/l)	73±7.07	77.5±3.54	71±1.41

4.2 Experiment two

4.2.1 Histological studies of gonad to determine the effects of hormone

The two different hormones Flash and PG were trialed for artificial breeding. After hormone trial, injected fishes both males and females were found not responded. Then injected brood fishes were sacrificed to determine the effectiveness of applied hormones. After histopathological studies shown that T₂ with Flash hormone was highly effective for gonadal maturation.

4.2.2 Histological observation of gonad

4.2.2.1 Stages of oocytes

Histological study of ovary for both Flash and PG hormone indicated the presence of four developmental stages: a) Early perinucleolar stage oocytes b) Late perinucleolar stage oocytes c) Yolk vesicle stage oocytes d) Yolk granular stage oocytes (Mature).

Early yolk granule stage (EYG): The yolk granules are formed by the fusion of small, coated vesicles which first appear peripherally in the oocyte. They stained light pink with haematoxylin and eosin.

Late yolk granule stage (LYG): During the late yolk granule stage, the yolk globules fuse with each other to form a single mass of yolk. The diameter of the oocytes increased simultaneously with the advancement of yolk granule stage and oil droplets appeared within the cytoplasm. The yolk granules appeared deep pink with haematoxylin and eosin.

Yolk vesicle stage (YV): The Yolk vesicle stage was characterized by the formation of yolk vesicles in the periphery of the oocytes. Initially they were formed as a single row which appeared colorless when the slides were stained with haematoxylin and eosin. These yolk vesicles developed as minute bodies but gradually increased in size and number. The nucleoli were usually present at the periphery of the nucleus, but in some cases, they also appeared elsewhere in the nucleus.

Pre-mature stage (PM): After the nucleus migration, nuclear membrane broke down. Yolk globules coalesced and no nucleus was observed, although the follicle layer was still visible.

Mature stage (M): The fully grown oocytes possess a large nucleus (germinal vesicle) in meiotic prophase. The germinal vesicle of this stage is located centrally or halfway between

the center and the oocyte periphery. Complete vision of yolk Globules was seen together with an overall increase in oocyte translucency.

In the present study, the four stages of oocytes were observed in different hormone treated ovary.

Treatment 1 with PG

Early stage oocytes were mostly evident. OG (oogonia), PNO, YV, EYG and Ch were observed in the histo-sections under microscope but few in number. PNO were high in number (Plate1).

Treatment 2 with PG

OG, PNO, YV and EYG were observed in the histo-sections under microscope. PNO was few in number; whereas YV oocytes were high in number (Plate1). PM and M stages were observed in few number.

Treatment 3 with PG

PNO, YV, EYG and LYG were observed in the histo-sections under microscope. YG (yolk granule) oocytes were highest in number although there were few PNOs (Plate1).

The difference among treatment were visual and clear. The doses of PG hormone were not very effective for inducing breeding because development of mature stages was lower in number. Among three treatments with PG were present of four stages (primary, yolk vesicle, vitellogenic and mature oocytes) but few in number. Besides, Flash hormone treated fish ovaries stages were comparatively having all stages but maturing stages higher in number.

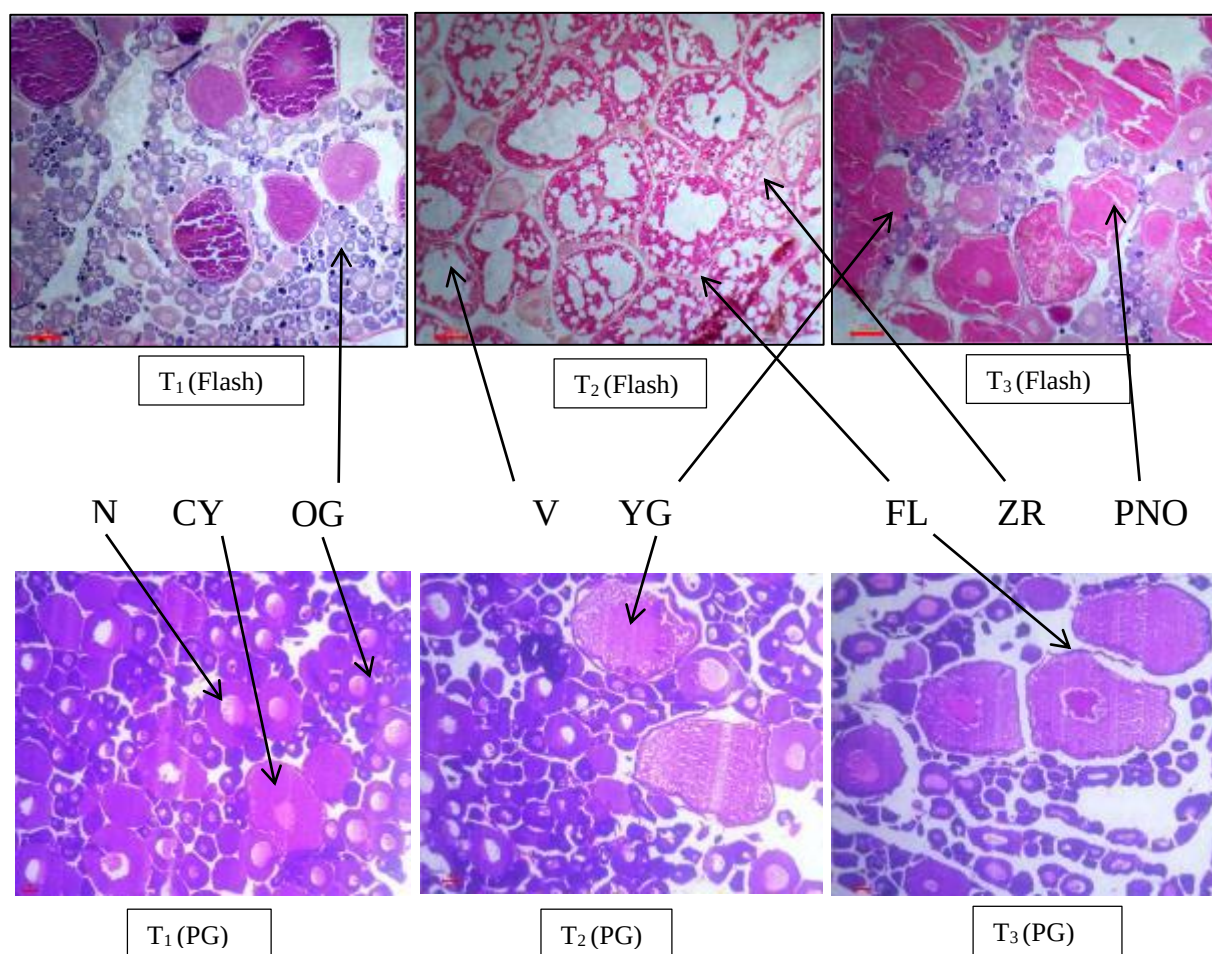


Plate 1. Ovary histological observations of different stages of *S. aor* oogenesis.

N: Nucleus, OG: Oogonium, CY: Cytoplasm, ZR: Zona radiata, CH: Chorion, YG: Yolk globule, V: Vacuole, FL: Follicular layer, PNO, perinucleolar oocyte.

4.2.3 Histological observation of ovary for T₂ with FLASH hormone

In treatment 2 with flash, having greater number of fully development stages. Eventually, the oocytes reached the late maturation stage when the envelope of the germinal vesicle disintegrated and the nuclear content mixed with the surrounding cytoplasm (Plate 1 (e)). At this stage, ovaries were large with a thin and transparent membrane. Oocytes were orange in colour, and peripheral and central blood vessels were evident. It is the proper time for spawning after hormonal induction.

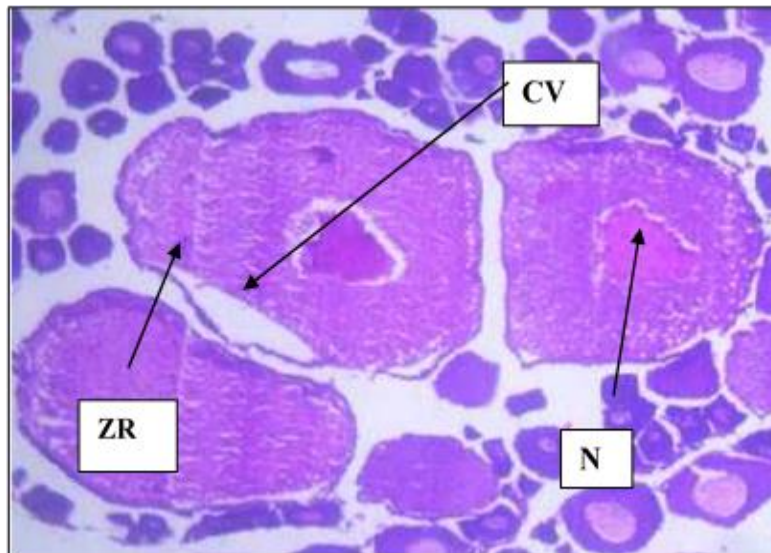
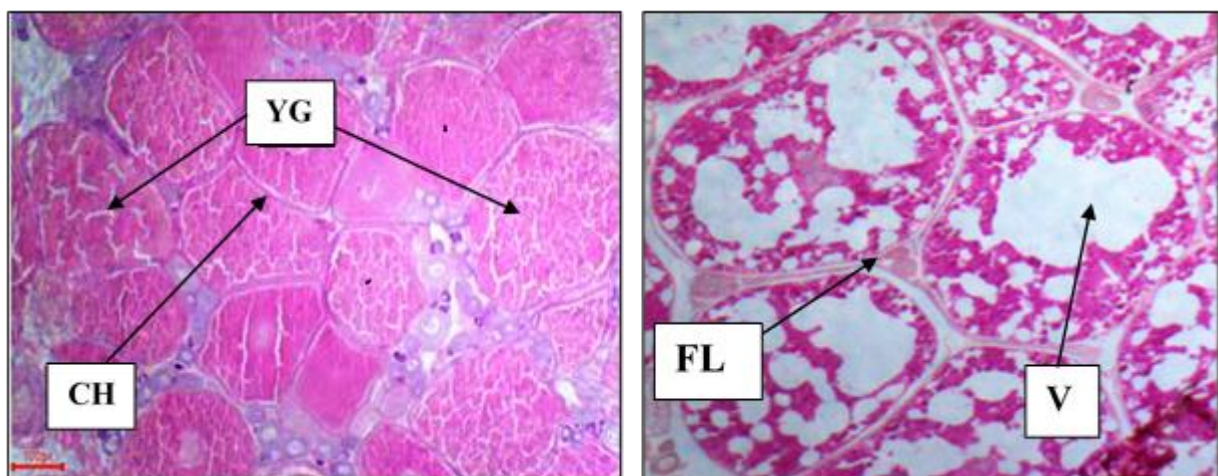


Plate 2. Vitellogenic growth phase (maturing): Primary and Secondary yolk globule oocyte of *S. aor* (H&E $\times 40$).

Vitellogenesis stage: Maturation of the egg is a long process that involves complex physiological and biochemical changes. One important step, vitellogenesis is a process in which yolk proteins are produced in the liver, transported to the ovary and stored in the egg resulting in tremendous egg enlargement. The yolk is important as a source of nutrition for the developing embryo. In the beginning of vitellogenesis, appearance of yolk and fat vacuoles in the ooplasm.



N: Nucleus, ZR: Zona radiata, CH: Chorion, YG: Yolk globule, V: Vacuole, FL: Follicular layer.

Plate 3. Maturation growth phase (mature): Migratory nucleus oocytes and Hydrated oocyte of *S. aor* (H&E $\times 40$)

Germinal vesicle migration and germinal vesicle breakdown (GVBD) are critical. Before it migrates the germinal vesicle or nucleus is located at the center of the egg in an arrested stage of development. At this stage, the egg is physiologically and genetically incapable of being fertilized even though it has the outward appearance of a fully mature egg. When conditions are appropriate for final maturation, nuclear development resumes and the germinal vesicle migrates to one side. Finally, the walls of the germinal vesicle break down, releasing the chromosomes into the cell. In mature eggs the migration of the germinal vesicle to the side of the cell will be completed. After the egg has matured, a class of compounds called prostaglandins is synthesized. These stimulate ovulation which is the rupture of the follicle cells that hold the egg. The egg is then released into the body cavity or ovarian lumen, where it may subsequently be released to the outside environment. Following ovulation, the viability of the eggs can decrease rapidly. Fish with gametes that have not yet been released by the gonads are called “green.” The term “ripe” is used to describe fish with gametes that have been released from the ovary into the ovarian lumen. Ripe fish can be stripped, green fish cannot). In the present study, brood fishes were stripped for releasing eggs but they were not released eggs. It might be in “green” condition. Some of the factors may be required to release the mature gametes into the lumen. Hence, they may release the eggs. In this experiment those required factors might be absent that is why brood fishes did not release eggs.

Thus, based on histological examination of the oocytes which showed the presence of up to four stages of oocyte development at various sampling times, it can be concluded that the mode of ovarian development of *S. aor* is asynchronous. In aquaculture, asynchronous oocyte development has practical importance in establishing a breeding programme as it indicates a consistent supply of eggs to the hatchery.

4.2.3 Stages of testicular germ cells

Histological studies of the testes showed that the course of spermatogenesis and the gradual changes in its development. The difference among treatment were visual and clear. The doses of PG hormone were not very effective for inducing breeding because development of mature stages was lower in number. Among three treatments with PG were present of stages ((spermatogonia (S), spermatocyte (SC), spermatozoa (SZ)) but few in number. Besides, Flash hormone treated fish testes stages were comparatively having all stages but maturing stages higher in number.

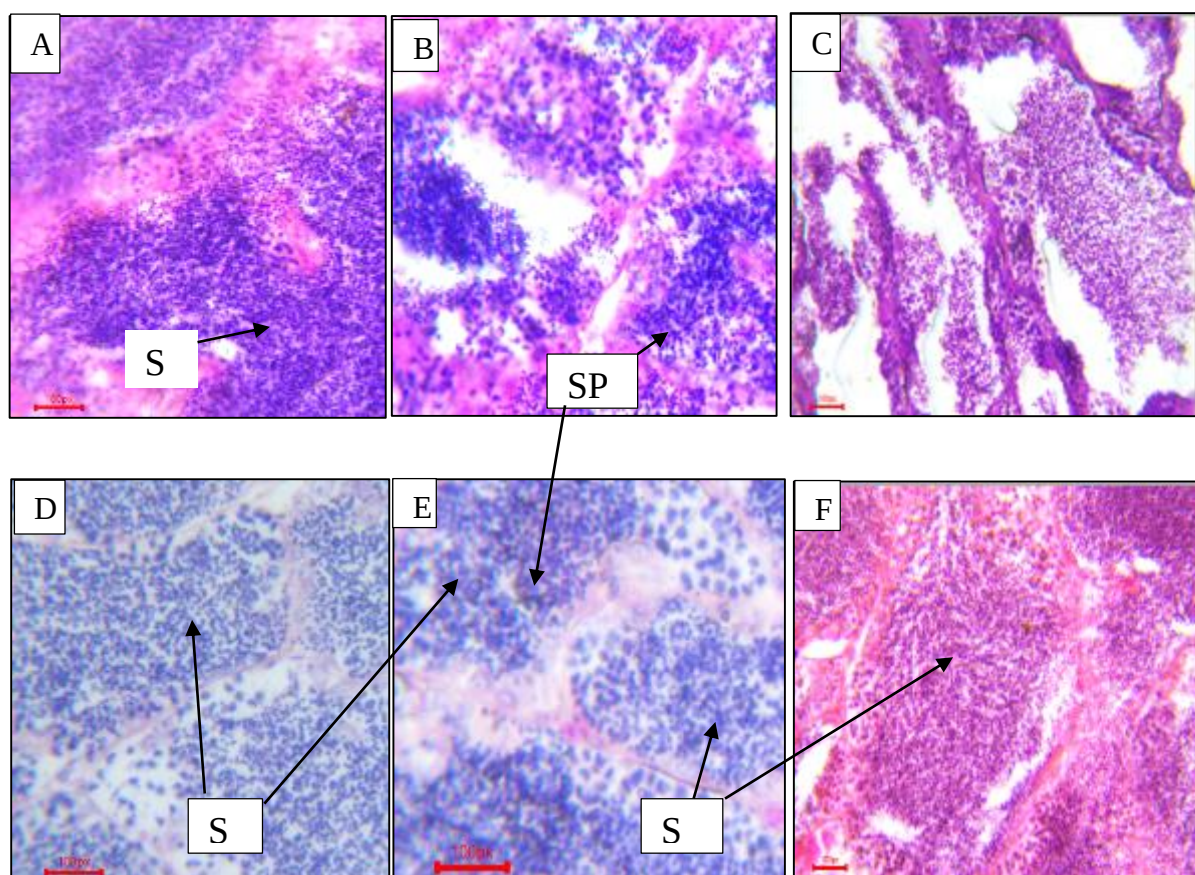


Plate 4. Photomicrograph of a cross section (CS) in testes at immature, mature stages. Here, A(T₁), B(T₂), C(T₃) indicated treated with PG and D(T₁), E(T₂), F(T₃) treated with Flash hormone.

4.2.3.1 Histological observation of testes for T₂ with Flash hormone

The maturing testis sets the beginning of active spermatogenesis. Nests of spermatogonia, spermatocytes and a few nests of spermatids can be detected. Concerning the spermatocytes, neither the nucleoli nor the cell membrane can be differentiated using the light microscope. The diameters and the comparatively little cytoplasm helped in distinguishing the primary and secondary spermatocytes. Sperm duct (SD) visible but thin. Spermatocytes (SC) more in number with Spermatogonia (S). Spermatozoa (SZ) in the sperm duct (SD) and sperm duct is thicker or fuller due to the spermatozoa and fluids in them. From the histological study of testis, the testicular germ cell stages of spermatogonia (S), spermatocyte (SC), spermatozoa (SZ) were observed. These findings agreed with the findings of Khan and Mollah (1998) for catfish *Pangasius pangasius*, Manosroi et al., (2004) for catfish *P. hypophthalmus*, Alam (2009) for the catfish *Ompok pabo* and Hassan (2006) who worked with the catfish *P. Nasutu*.

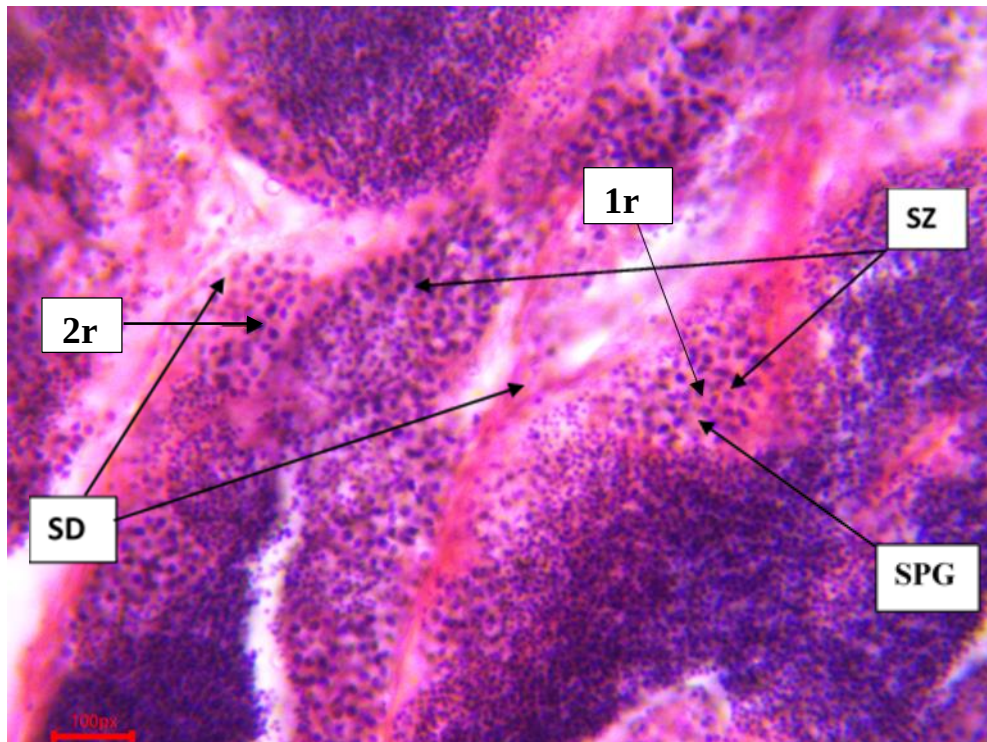


Plate 5. Photomicrograph of a cross section in maturing testes showing few nestes of spermatids (SD) and large number of spermatocytes nests (1ry & 2ry) and spermatogonia (spg). (H & E). x 40.

CHAPTER FIVE

SUMMARY

S. aor locally called Ayre is a riverine catfish dominant in the greater haor region of Sylhet and other inland water bodies of Bangladesh. It is considering highly delicious food item for the domestic market in our country. Gonadal developments were examined by macroscopically and microscopically and histological observations were assessed to establish the oocyte development of ovary. Macroscopically four stages of oocyte development were identified and which is further demonstrated by histological study. Regarding the information available on different aspects of reproductive biology, except proper documentation on spatial variation in breeding periodicity, not much information is available on other aspects like sex-ratio, size at first maturity and fecundity. Spatial variation in breeding periodicity has been reported for *S. aor* and different factors like temperature, day length, water current, sandy bed and hydrological parameters have been attributed for such variation. Information so far available on sex-ratio is quite contradictory. Ramakrishniah (1992) and Saigal (1964) reported female dominance while Azadi et al. (1992) have reported male dominance in *S. aor* population. Therefore, detail work is needed to get firm conclusion in this aspect. Information available on size at first maturity is also scanty as that of fecundity. Fecundity ranges reported by Azadi et al. (1992) and Ranganathan and Radha (1966) have high difference in respect to the information provided by Saigal (1964). Therefore, further study is also needed for these aspects. This fish species is much hardy in nature and it can tolerate great extent of odd environmental condition when a few carps will survive (Chondar 1999). So it can be a good species to culture in captivity. Until now fishery of *S. aor* mainly depends on capture from nature. For fishery, however, captive culture is needed which until now has not been tried at all. Therefore, this study was conducted to gather important information needed for its captive culture to explore its future fishery. In treatment 1, Two tanks have total 16 fishes. After 3 weeks later, 15 fishes were found dead. only one infected fish was alive. The average percentage of survival rate was 6.25 in cemented tanks (Table 8). It might be due to physical injury which occurred from frequent contact with hard bottom surface of the tanks. Because all the dead fishes seriously injured on abdominal sides. Besides, the skin and barbells of fishes were observed broken followed by gradual infections and lesions and thus higher mortality occurred. After completion of treatment 1, tanks were pre-arranged having sides pasted with cement and bottom filled with mud. A total of 16 fish were releases in two tanks each tank had 8 fishes. Fishes were routinely monitored to observe the condition of brood

fishes. After four weeks of stocking, it was observed only two brood fishes survived in each tank and other brood fishes found deceased. The frequent contact with hard surface of cemented tank resulted lesions in fish body and turbidity caused oxygen deficiency might be accountable for the mortality in this treatment. The average percentage of survival rate was found only 25 in cemented tanks with muddy bottom layer (Table 9). In treatment 3, tanks condition was again modified. Here bottom of the tanks was filled with sands. Five weeks later, all brood fishes of tank 1 found alive with healthy condition, meanwhile, only one brood fish of tank 2 was found dead. Survival rate of brood fish were observed 100% and 87.5% in tank 1 and tank 2, respectively. While, cumulative survival rate of treatment 3 was estimated 93.75% (Table 10). Sandy bottom of the experimental tanks might be the reason of low mortalities in the current treatment. These findings are in agreement with the findings of Ranganathan and Radha (1966) and Saigal (1982). They reported *S. aor* preferred sandy beds and clear water for living. Finally, found suitable condition and maximum survival rate of *s. aor* in cemented tanks with sandy bottom layer. The ovary of the *S. aor* was paired, more or less tubular structure lying dorsal to the alimentary canal and ventral to the swim bladder. On the other hand, the testis was bi-lobed and more or less elongated. The anterior region was wider and containing more projections than the posterior. The posterior tip was projected towards the tail region. Histological study of gonad helped to determine the maturation stages of both male and female. It was full development stages in both ovaries and testis found in Flash hormone treatment 2. After histological observation found Flash hormone in treatment 2 was highly effective in gonadal maturation than PG. To observe the effective dose for induced breeding, the females were first injected at the rate of 0.5 (T₁), 1.0 (T₂), and 1.5 (T₃) ml Flash/kg body wt. while the second doses were performed at the rate of 1.0 (T₁), 1.5 (T₂), and 2.0 (T₃), ml Flash /kg body wt. after 6 hrs. from the first dose. On the other hand, the males were administered a single dose of 1 ml FLASH /kg body wt. at the time of second dose of the female. Consequently, the females were first injected at the rate of 1.0 (T₁), 1.5 (T₂), and 2.0 (T₃) mg PG/kg body wt. while the second doses were performed at the rate of 2.0 (T₁), 2.5 (T₂), and 3.0 (T₃), mg PG /kg body wt. after 6 h from the first dose. On the other hand, the males were administered a single dose of 2 mg PG /kg body wt. at the time of second dose of the female. Flash hormone was found to be effective hormone to maximize the ovulation. None of the trialed fishes released any eggs nor do any sperm by 72 hrs after injection with different doses. Then the injected brood fishes were sacrificed to know the effectiveness of two different hormones. Based on the results of ovarian histological analysis, *S. aor* has group asynchronous mode of ovarian development.

Therefore, it is capable of spawning many times in a year outside the peak spawning season, when spawning conditions are favorable. Germinal vesicle migration and germinal vesicle breakdown (GVBD) are critical. Before it migrates the germinal vesicle or nucleus is located at the center of the egg in an arrested stage of development. At this stage, the egg is physiologically and genetically incapable of being fertilized even though it has the outward appearance of a fully mature egg. When conditions are appropriate for final maturation, nuclear development resumes and the germinal vesicle migrates to one side. Finally, the walls of the germinal vesicle break down, releasing the chromosomes into the cell. In mature eggs the migration of the germinal vesicle to the side of the cell will be completed. After the egg has matured, a class of compounds called prostaglandins is synthesized. These stimulate ovulation which is the rupture of the follicle cells that hold the egg. The egg is then released into the body cavity or ovarian lumen, where it may subsequently be released to the outside environment. Following ovulation, the viability of the eggs can decrease rapidly. Fish with gametes that have not yet been released by the gonads are called “green.” The term “ripe” is used to describe fish with gametes that have been released from the ovary into the ovarian lumen. Ripe fish can be stripped, green fish cannot. In the present study, all eggs were found fully matured but they were in “green” condition. Some of the factors may be required to release the mature gametes into the lumen. Hence, they may release the eggs. In this experiment those required factors might be absent that is why brood fishes did not release eggs. Again microscopic observations of testicular germ cells revealed presence of spermatocyte (SC), spermatid (ST) and spermatozoa (SZ) stages. Presence of mature testicular germ cells in high number inside the sperm duct samples indicates period of testicular maturity of male *S. aor*.

CHAPTER SIX

CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

Many species of fish will not readily reproduce under certain culture conditions. Others will but not necessarily when the farmer desires. In these cases, induction of spawning can be of great value. Two techniques are commonly used, sometimes in conjunction with one another. The first is manipulation of the culture environment to mimic some important quality in the fish's natural environment. The second is injection of hormones to stimulate spawning. The hormones may be natural hormones taken from fish or other animals, genetically engineered from bacteria or synthetic analogs of naturally occurring hormones. Methods vary from species to species and situation to situation. However, at least two generalizations can be drawn. First, brooders are very vulnerable to rough handling. Care should always be used to avoid damaging these valuable animals. Second, a fish that does not have mature gametes will not produce viable eggs or sperm no matter how many times it is injected with hormones. Ripeness is the result of environmental factors working over a period of time leading to maturation of the gonads and production of viable eggs. Many procedures have been developed for inducing fish to undergo the last steps of spawning. Farmers should thoroughly research the procedures that have been developed for their species of fish through experimentation and select those that best suit the circumstances. In addition, once the fish have spawned, there are many techniques involved in incubating and caring for the eggs and caring for the hatched fry. These too must be thoroughly researched. It is a popular food fish due to its good taste and high nutritional value. Recently it has also made its entry in domestic ornamental fish markets of India and has been reported to have moderate export price too. Earlier, few works on different aspects of its morphology and biology have been carried out but no consolidated review is available on these aspects. Therefore, the current work was done to sum up all available information on these aspects to fill the information gap that will be beneficial to its future fishery. Considering all available information, knowledge on food and feeding habit of this fish species is satisfactory, except detail information on spatial variation of its breeding periodicity, proper information on other aspects are still lacking. No work has been conducted to examine its captive culture potential or for induced breeding. Further work should be done to explore its future fishery. Information on optimum condition for survival rate and effective doses is important for planning, conservation and improvement of artificial breeding of a threatened fish species. The results of the study would be effective tools towards designing management and conservation

plans for sustainable conservation and captive maturation. And used cemented tank with sandy bottom for getting maximum survival rate and more profit. The information gathered through the present experiment may be used for better management of Ayre catfish fishery in this country towards conservation and for induction of a potential species into aquaculture.

6.2 Recommendations

The cat fish resources are an important aspect of fishery potential of a water body. Catfishes due to their great demand and high market value and non-availability of stocking material largely hinder the organized culture of these fishes in our country. The small catfish species are noteworthy for their size, taste and market value. They are distributed in lentic and lotic water bodies and breed naturally in perennial rivers during monsoon. Though breeding and larval rearing of certain catfishes has been done successfully for their commercial production is yet to be achieved. Hence, maintenance of catfishes plays a key role in achieving successful induced breeding. Therefore, further study is also needed for these aspects. This fish species is much hardy in nature and it can tolerate great extent of odd environmental condition when a few carps will survive (Chondar 1999). So it can be a good species to culture in captivity. Until now fishery of *S. aor* mainly depends on capture from nature. For fishery, however, captive breeding is needed which until now has not been tried at all. Therefore, proper study should be conducted to gather important information needed for its induced breeding to explore its future fishery.

On the basis of information obtained from this study, the following recommendations may be drowning:

- ❖ More detailed studies are needed for induced breeding and development of fry.
- ❖ Studies are needed to culture the species in captive condition.
- ❖ Using cemented tank with sandy bottom for getting maximum survival rate and more profit in artificial condition.

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APPENDICES

Appendix 1

Monthly used brood fishes for improvement of survival rate and artificial breeding of *S. aor*

Months	Sample No	Body Wt.	Standard Wt.
May	1	1094.24	38.10
May	2	702.56	32.20
May	3	768.43	34.30
May	4	705.34	30.40
May	5	654.31	29.50
May	6	568.98	28.80
May	7	831.1	36.10
May	8	718.79	35.20

Months	Sample No	Body Wt.	Standard Wt.
June	1	718.79	31.75
June	2	651.83	25.35
June	3	724.45	27.95
June	4	695.52	31.64
June	5	771.43	32.38
June	6	713.34	33.17
June	7	689.67	35.06
June	8	821.66	33.40

Months	Sample No	Body Wt.	Standard Wt.
July	1	691.48	32.38
July	2	743.13	33.17
July	3	685.57	35.06
July	4	634.65	33.40
July	5	747.24	27.72
July	6	623.11	28.64
July	7	493.78	31.85
July	8	562.18	29.34

Months	Sample No	Body Wt.	Standard Wt.
August	1	775.31	33.20
August	2	712.83	33.30
August	3	756.42	33.40
August	4	784.15	33.45
August	5	779.28	33.50
August	6	889.54	36.10
August	7	898.16	36.20
August	8	913.23	37.30

Months	Sample No	Body Wt.	Standard Wt.
September	1	775.31	33.20
September	2	712.83	33.30
September	3	756.42	33.40
September	4	784.15	33.45
September	5	779.28	33.50
September	6	889.54	36.10
September	7	898.16	36.20
September	8	913.23	37.30

Appendix 2

Estimation of Brood fish's survival rate in cemented tank

Tank No.	No. of Fish Release	No. of dead fish/Week			No. of Live Fish	Survival Rate (%) / Tank	Average Survival rate (%)
		1st	2nd	3rd			
1	8	0	2	5	1	12.5	6.25
2	8	1	3	3	0	0	

Appendix 3

Estimation of Brood fish's survival rate in cemented tank with muddy bottom

Tank No.	No. of Fish Release	No. of dead fish/Week				No. of Live Fish	Survival Rate (%) / Tank	Average Survival rate (%)
		1st	2nd	3rd	4th			
1	8	0	2	3	1	2	25	25
2	8	0	3	2	1	2	25	

Appendix 4.

Estimation of Brood fish's survival rate in cemented tank with sandy bottom

Tank No.	No. of Fish Release	No. of dead fish/Week					No. of Live Fish	Survival Rate (%) / Tank	Average Survival rate (%)
		1st	2nd	3rd	4th	5th			
1	8	0	0	0	0	0	8	100	93.75
2	8	0	1	0	0	0	7	87.5	

Appendix 5

Water quality parameters

Water Quality Parameters	Treatment One		Treatment Two		Treatment Three	
	Tank 1	Tank 2	Tank 1	Tank 2	Tank 1	Tank 1
DO (mg/l)	8	7	7	8	8	7
pH	6.8	7.0	7.2	7.2	7.6	7.4
Temperature (°C)	28	27	28	30	27	28
Turbidity (cm)	18	22	26	28	10	9
Hardness	78	89	74	69	68	70
Nitrite (mg/l)	0.04	0.05	0.03	0.04	0	0.01
Ammonia (mg/l)	0.14	0.18	.017	0.019	0.021	0.011
Alkalinity (mg/l)	78	68	80	75	72	70

